

Lecture 5

Eigenvalues and Eigenvectors

Throughout, the core uses a nonzero finite-dimensional V over the field stated in the problem, $T \in \mathcal{L}(V)$ is an operator, and $E(\lambda, T) = \text{null}(T - \lambda I)$ is its eigenspace. Here “spectrum” means the finite point spectrum; the general Hilbert-space spectrum is introduced in Lecture 14. Tags mark the flavour of each problem; a ★ marks a harder one.

Core tutorial route

Work **5.1, 5.2, 5.7, 5.14, 5.18, 5.19, 5.20, and 5.21** in that order (about 80–100 minutes before discussion). This eight-problem route checks eigendata and characteristic invariants, real/complex transfer, an eigenspace proof, invariant decomposition and classification, reconstruction of the finite complex existence argument, and a constructive triangularizing flag.

Additional practice: 5.4–5.5, 5.9–5.13, 5.16–5.17, 5.22, 5.24. **Bridge:** 5.3, 5.6 (calculus and complex-field applications). **Extension:** 5.8, 5.15, 5.23 (harder counting/product proofs and generalized eigenvectors). Generalized-eigenvector work is not core progress.

Finding eigenvalues and eigenvectors

Exercise 5.1 (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$. Recover $\text{tr } A$ and $\det A$ from the eigenvalues and check them directly.

Answer. Eigenvalues 2, -1 ; $E(2, A) = \text{span}((5, 2))$, $E(-1, A) = \text{span}((1, 1))$; $\text{tr } A = 1$, $\det A = -2$.

Exercise 5.2 (computational) Compute the eigenvalues of the rotation matrix $\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$ over $\mathbb{C}$. For which α are they real?

Answer. $\lambda = e^{\pm i\alpha}$; real iff $\sin \alpha = 0$, i.e. $\alpha \in \pi\mathbb{Z}$ (within $[0, 2\pi)$, just $\alpha = 0, \pi$).

Exercise 5.3 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be differentiation. Find all eigenvalues and eigenvectors of D .

Answer. Only eigenvalue 0; $E(0, D) = \text{span}(1)$ (geometric multiplicity 1, so D is defective).

Exercise 5.4 (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

Answer. Eigenvalues 3, -1 ; $E(3, A) = \text{span}((1, 1))$, $E(-1, A) = \text{span}((1, -1))$.

Exercise 5.5 (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix}$.

Answer. Eigenvalues 2, 3; $E(2, A) = \text{span}((1, 2))$, $E(3, A) = \text{span}((1, 3))$.

Exercise 5.6 (computational) Working over $\mathbb{C}$, find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$.

Answer. Eigenvalues $\pm i$; the eigenspaces are $E(i, A) = \text{span}((1 + i, 1))$ and $E(-i, A) = \text{span}((1 - i, 1))$.

Exercise 5.7 (proof) Suppose T is invertible and λ is an eigenvalue of T . Prove that $\lambda \neq 0$ and that $1/\lambda$ is an eigenvalue of T^{-1} , with the same eigenvectors.

Exercise 5.8 (★) Extension. Let J be the $n \times n$ real matrix with every entry equal to 1, where $n \geq 2$. Find all eigenvalues of J , their algebraic and geometric multiplicities, and a basis of each eigenspace.

Hint. J has rank 1; look at the all-ones vector and at the vectors whose entries sum to 0.

Structure

Exercise 5.9 (computational) Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Find the eigenvalues of $A^2 - A$.

Answer. 0 and 6.

Exercise 5.10 (computational) Verify that $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is an involution and find $E(1, A)$ and $E(-1, A)$.

Answer. $A^2 = I$; $E(1, A) = \text{span}((1, 1))$, $E(-1, A) = \text{span}((1, -1))$.

Exercise 5.11 (computational) The matrix $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ is invertible. Find the eigenvalues of A^{-1} together with a corresponding eigenvector for each.

Answer. Eigenvalues $\frac{1}{2}$ (eigenvector $(1, -1)$) and $\frac{1}{3}$ (eigenvector $(0, 1)$).

Exercise 5.12 (proof) Suppose λ is an eigenvalue of T and $p \in \mathcal{P}(\mathbb{F})$. Prove that $p(\lambda)$ is an eigenvalue of $p(T)$, with the same eigenvector.

Exercise 5.13 (proof) For finite-dimensional V , prove that T is invertible if and only if 0 is not an eigenvalue of T .

Exercise 5.14 (proof) Suppose $T^2 = I$ (an involution). Prove that every eigenvalue of T is ± 1 , and that $V = E(1, T) \oplus E(-1, T)$.

Hint. For the eigenvalues, apply T twice to an eigenvector. For the splitting, decompose v into a part T fixes and a part T negates.

Exercise 5.15 (★) Extension. Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that ST and TS have the same eigenvalues.

Hint. If $STv = \lambda v$ with $\lambda \neq 0$, look at how TS acts on Tv . Treat $\lambda = 0$ separately through invertibility.

Exercise 5.16 (proof) Call T *nilpotent* if $T^k = 0$ for some $k \geq 1$. Prove that the only eigenvalue of a nilpotent operator is 0.

Upper-triangular form

Exercise 5.17 (computational) Read off the eigenvalues of $A = \begin{pmatrix} 5 & 2 & 7 \\ 0 & 5 & -1 \\ 0 & 0 & 3 \end{pmatrix}$ and state its trace and determinant. Is A invertible?

Answer. Eigenvalues 5, 5, 3; $\text{tr } A = 13$, $\det A = 75$; invertible.

Exercise 5.18 (proof) For $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, classify all invariant subspaces among

$$\begin{aligned} U_1 &= \text{span}((1, 0, 0)), & U_2 &= \text{span}((1, 1, 0)), \\ U_3 &= \text{span}((1, 0, 0), (0, 1, 0)), & U_4 &= \text{span}((0, 0, 1)). \end{aligned}$$

Identify an invariant subspace that is not merely an eigenline.

Answer. U_1, U_3, U_4 are invariant; U_2 is not. The plane U_3 is invariant but is not an eigenline.

Exercise 5.19 (proof) Let T act on a nonzero complex vector space of dimension n . Reconstruct the finite eigenvalue-existence proof by explaining why $v, Tv, \dots, T^n v$ is dependent, why the resulting annihilating polynomial has positive degree, where factorization over $\mathbb{C}$ is used, and why one linear factor must have nonzero kernel.

Exercise 5.20 (computational) For $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ acting on $\mathbb{C}^2$, construct an ordered basis in which A is upper triangular. Exhibit the corresponding invariant flag and verify the triangular matrix directly.

Answer. For example $v_1 = (1, i)$, $v_2 = (1, 0)$ gives $[A]_{(v_1, v_2)} = \begin{pmatrix} -i & -i \\ 0 & i \end{pmatrix}$; $0 \subset \text{span}(v_1) \subset \mathbb{C}^2$ is invariant.

Exercise 5.21 (proof) Suppose T has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$ with respect to a basis $v_1, \dots, v_n$. Prove that $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$.

Exercise 5.22 (★) Prove that if finite-dimensional $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable (its eigenvectors form a basis).

Hint. Eigenvectors belonging to distinct eigenvalues are independent; count them against $\dim V$.

Exercise 5.23 (computational) **Extension (generalized eigenvectors).** For $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ acting on $\mathbb{F}^2$, find the eigenspace $E(3, A)$ and the generalized eigenspace $G(3, A)$, and exhibit a Jordan chain.

Answer. $E(3, A) = \text{span}((1, 0))$ ($\dim 1$), $G(3, A) = \mathbb{F}^2$ ($\dim 2$); Jordan chain $(0, 1) \rightarrow (1, 0) \rightarrow 0$.

Exercise 5.24 (proof) Let $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Prove that T and $T + cI$ have exactly the same eigenvectors, and that λ is an eigenvalue of T if and only if $\lambda + c$ is an eigenvalue of $T + cI$.